

PayPal API Testing

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What is an API?



An Application Programming Interface acts as a messenger, allowing different software applications to communicate and share data

Think of it as a digital waiter taking your order (request) to the kitchen (server) and bringing back your meal (response).

Welcome Message (CEO)

Company Presentation

HTTP Methods



Get

Retrieve Data

Post

Create new Data

Put

Update existing Data

Delete

Remove Data

HTTP Status Codes —————→

2xx Success

Request received, understood, and
accepted

4xx Client Error

Client made an invalid request.

5xx Server Error

Server failed to fulfill a valid request.

Why Postman



User-Friendly Interface

- Intuitive design for sending requests
- Easy to inspect responses

Collection Organization

- Group requests logically
- Share with team members

Scripting & Automation

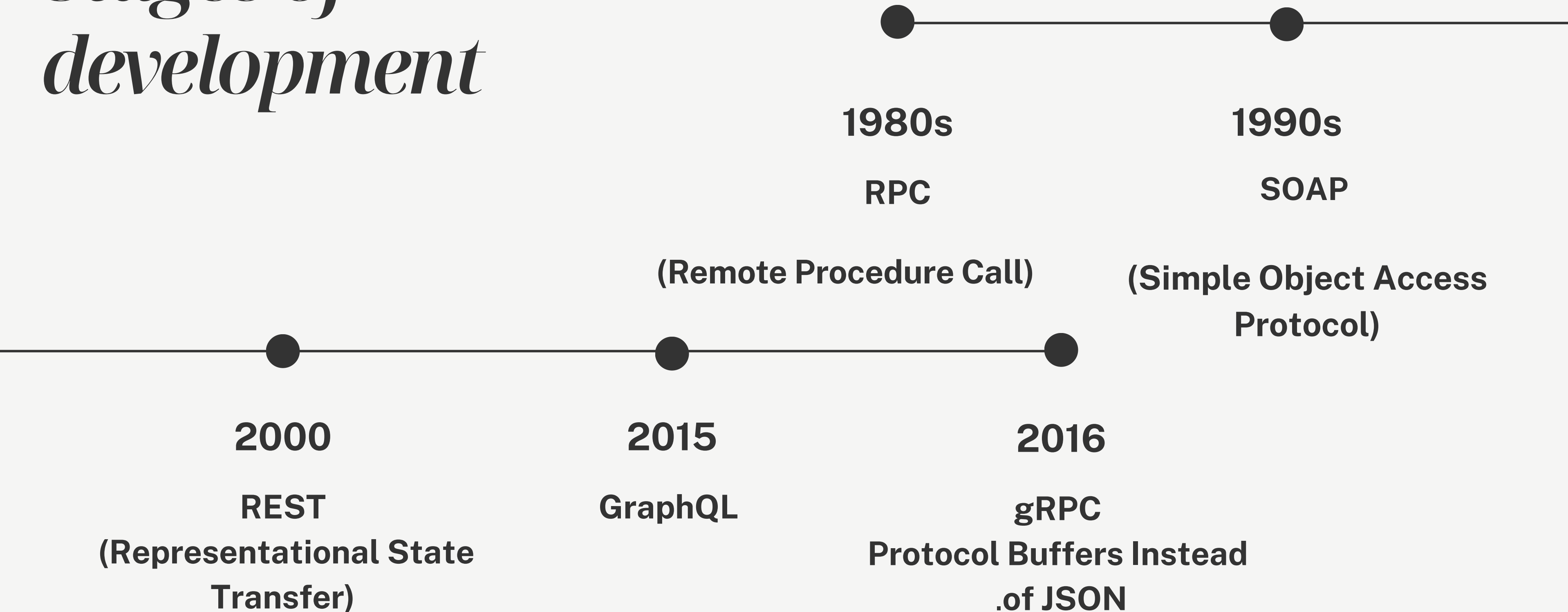
- Pre-request scripts for setup
- Test scripts for validation

Why RestAssured

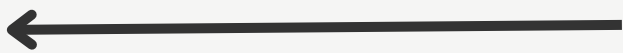


- Java Based
- API Automation
- TestNG Integration
- Assertions
- CI/CD Support

Stages of development



Technologies



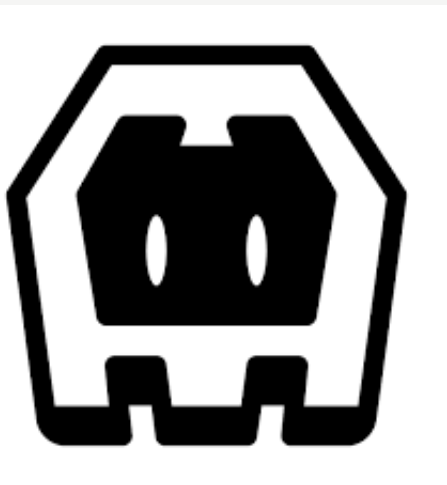
Java



Postman



GitHub



Maven

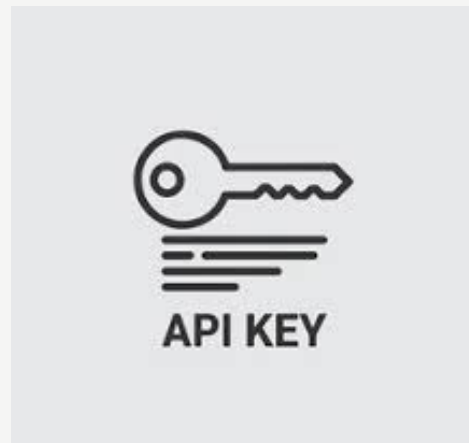


Rest Assured

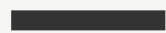


TestNG

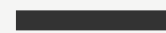
Challenges



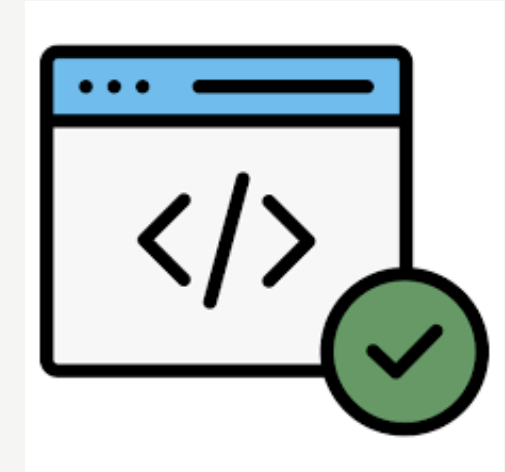
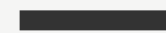
**OAuth
Authentication**



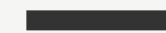
**Dynamic Access
Token**



**Environment
Variables**



**Response
Validation**



Manual Api

**Tested PayPal Authentication APIs manually using Postman.
Verified both positive and negative scenarios based on the expected results.**

The screenshot displays a Postman interface for testing the PayPal Authentication API. The request is a POST to the endpoint `{{base_url}}/v1/oauth2/token/terminate`. The body is set to `x-www-form-urlencoded` with the following parameters:

key	value	Description
☒ token	INVALID_TOKEN	
☒ token_type_hint	ACCESS_TOKEN	
☐ grant_type	client_credentials	
☐ grant_type	refresh_token	
☐ refresh_token	{{refresh_token}}	
☐ grant_type	authorization_code	
☐ code	{{authorization_code}}	
Key	Value	Description

The response is a `401 Unauthorized` status with a response time of 721 ms and a size of 1.23 KB. The response body is JSON:

```
1 {
2   "error": "invalid_token",
3   "error_description": "Access Token not found in cache"
4 }
```

GitHub

Source Code

Store the complete API automation project.

Organize Java classes, Maven files, and TestNG suites.

Version Control

Track all code changes.

Restore previous versions when needed.

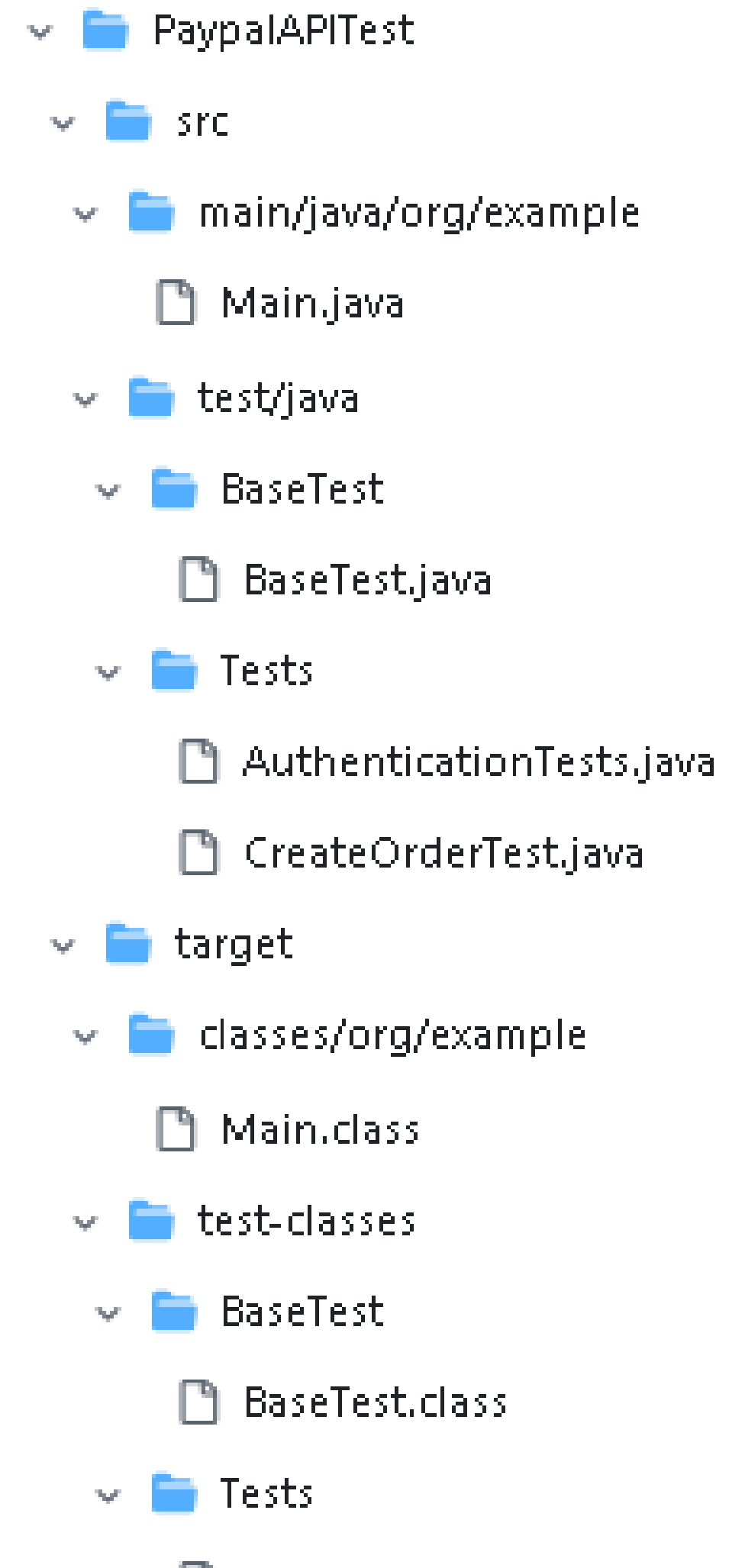
Manage project history.

Collaboration

Share the project with team members.

Review code changes.

Work together using GitHub repositories.



API Authentication Methods

How clients prove who they are to an API

API Key

A unique secret key sent in a header or query parameter to identify the calling application.

Basic Auth

Base64-encoded username and password passed in the Authorization header over HTTPS.

Bearer Token

A token (often a JWT) sent as 'Authorization: Bearer <token>' to access protected resources.

OAuth 2.0

Token-based delegated access; common flows include client-credentials and authorization-code.

Types of API Testing

1

Functional

Verifies endpoints return correct results for valid requests.

2

Negative

Confirms the API rejects invalid input with proper errors.

3

Security

Checks authentication, authorization and injection (SQLi / XSS).

4

Performance

Measures response time and behaviour under load.

5

Integration

Validates business flows across multiple endpoints.

REST vs SOAP vs GraphQL

	REST	SOAP	GraphQL
Format	JSON over HTTP	XML, strict schema	Query language, JSON
Transport	HTTP	HTTP / SMTP	HTTP (single endpoint)
Data fetching	Multiple endpoints	Defined operations	Client picks exact fields
Flexibility	Moderate	Rigid / contract-based	High
Best for	Most web & mobile APIs	Enterprise / banking	Tailored client needs

Postman vs Rest Assured

Postman

- ✓ GUI tool — fast, exploratory testing
- ✓ Organize requests into collections
- ✓ Scripting & assertions in JavaScript
- ✓ Great for manual & light automation
- ✓ Runs in CI via the Newman CLI

Rest Assured

- ✓ Java library — code-based automation
- ✓ Integrates with TestNG / JUnit + Maven
- ✓ Tests are version-controlled with the app
- ✓ Strong for regression & large suites
- ✓ First-class fit for CI/CD pipelines

API Testing Best Practices

- ✔ Test both positive and negative scenarios
- ✔ Externalize configuration & secrets (no hardcoding)
- ✔ Validate status code, schema, body & response time
- ✔ Use data-driven testing to widen coverage
- ✔ Keep tests independent and repeatable
- ✔ Automate execution inside CI/CD pipelines

Thank you

For your attention

